SOLUTION OF AC SOURCE DRIVEN LCR-SERIES CIRCUIT FOR OBSERVATION OF COMPLETE RESONANCE

A Term Paper
Submitted To Patan Multiple Campus, IOST TU, Patandhoka, Lalitpur,
In the Partial Fulfillment for The
Requirement Of Master's Degree of Science in Physics

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Symbol No.: 3683/076
June, 2023

Recommendation

It is certified that Mr. **Ram Chandra Giri** Has carried out the term-paper entitle **"SOLUTION OF AC SOURCE DRIVEN LCR-SERIES CIRCUIT FOR OBSERVATION OF COMPLETE RESONANCE",** under our supervision. We recommend the dissertation submitted in partial fulfilment for the requirement of master degree of science in Physics at Tribhuvan University.

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Acknowledgements

I am grateful to the many people for their time, support and encouragement.

First and foremost, I would like to thank my advisor, Associate Prof. Suresh Prasad Gupta. His insight and guidance have been of great help during my work on this paper. His advice will benefit me the rest of my life.

I would also like to thank the members of evaluation committee, for taking the time to review my work. Their time and input will be greatly appreciated.

I thank Rajesh Kumar Bachhan Sir & Prakash Man Shrestha Sir for all their advice. I give my gratitude for providing support during my work. Patan Multiple Campus, Lalitpur for the academic environment. Also, I am thankful to all the faculty member of the department for their suggestion and cooperation throughout my master's degree.

I must also thank my many friends for their support and friendships during my time at PMC. In particular, I thank Sanjay Karki & Sukudev Rijal for being with me together on PMC. I also would like to give my thanks to Bhuwan Rana for his support on this paper. I wish all my friends the best.

Finally, I give my thanks to my family, who were always there to support me in mind and spirit.

Evaluation

The undersigned that we read, approved and recommended for term paper of Master's in Physics, Patan Multiple Campus, Patandhoka Lalitpur, Tribhuvan University, Nepal for accepting the term paper entitled "SOLUTION OF AC SOURCE DRIVEN LCR-SERIES CIRCUIT FOR OBSERVATION OF COMPLETE RESONANCE", by Mr. Ram Chandra Giri.

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Abstract

In this paper, complete response of a series LCR circuit are observed using the Runge-Kutta 4th order method. Kirchhoff's law was used and second order differential equation was formulated. This second order differential was transformed into 1st order differential equation by substitution. Using FORTRAN together with gnuPlot, we observed the complete response in the different numerical cases (same AC source but different resistance values). Also analyzed the absolute error between analytically solved and numerical method solved case.

For a particular example. we observed the Complete response with variations in the value of resistance R. From the observation, we found the current to propagate through a sinusoidal pattern after initial spike due to AC source. How fast it settles into a sinusoidal pattern depends on the value of resistance, that is transient case. Absolute error between calculated current using Analytical method and RK4 method is found to be in the order of 10^{-6} or less.

Keywords: LCR circuit, RK4 method, Numerical Method, 2nd order differential equation

Contents:

Recommendation.....i
Acknowledgments.....ii
Evaluation.....iii
Abstract.....iv
Contents……v
Table of Figures.....vi
Chapter 1: Background.....1
Chapter 2: Objectives.....5
Chapter 3: Literature Review.....6
Chapter 4: Theoretical Framework.....7
Chapter 5: Data Set and Methodology.....11
Chapter 6: Result and Discussion.....19
Chapter 7: Conclusion.....22
Chapter 8: Future Works.....24
References.....25
Appendix.....26

Table of Figures:

Figure 1: RLC series circuit [11] ……...…1

Figure 2: Plot showing underdamped and overdamped responses of a series RLC circuit to a voltage input step of 1 V. The critical damping plot is the bold red curve. The plots are normalized for L = 1, C = 1 and $\omega_0 = 1$. [12] ……..4

Figure 3: Graph of natural response for analytical solution, underdamped case ……13

Figure 4: Graph of natural response with RK4 method, underdamped case ……...14

Figure 5: Graph of Forced response obtained through RK4 method, underdamped case …...14

Figure 6: Graph of natural response with RK4 method, critical damping case ……..16

Figure 7: Graph of forced response with RK4 method, critical damping case ……..….16

Figure 8: Graph of natural response with RK4 method, overdamping case ……...…17

Figure 9: Graph of forced response with RK4 method, overdamping case ……....18

Figure 10: Plot of current data for natural response of both analytical and numerical methods, numerical case 1 ……..19

Figure 11: Absolute error V/S Time using data of analytic current and numerical method current for numerical case 1 ……19

Figure 12: Complete response as a sum of Normal response and Forced response for numerical case 1 ……..……20

Figure 13: Complete response as a sum of Normal response and Forced response for numerical case 2 ……..20

Figure 14: Complete response as a sum of Normal response and Forced response for numerical case 3 …….…….21

Figure 15: Completer response numerical cases 1, 2, 3 …….….22

Figure 16: Completer response numerical cases 1, 2, 3 with limit lower than figure 15 ……22

Chapter 1: Introduction / Background:

A circuit consisting of a resistor (R), and inductor (L), and a capacitor (C), which is either connected in series or in parallel is RLC circuit. Sequence of the constituent components may vary for different circuit combination.

The circuit forms a harmonic oscillator for current, and resonates like LC circuit. While introduction of the resistor in the circuit causes damping. But, even without resistor there will be some resistance in the circuit.

RLC circuits have many applications as oscillator circuits. As a tuned circuit, Radio receivers and television sets use them for tuning to select a narrow frequency range from ambient radio waves. An RLC circuit can be used as a band-pass filter, band-stop filter, low-pass filter or high-pass filter. The tuning application, for instance, is an example of band-pass filtering. The RLC circuit can be described by a second-order differential equation in circuit analysis for any voltage or current in the circuit. [1]

We will take a closer look at the Series LCR circuit.

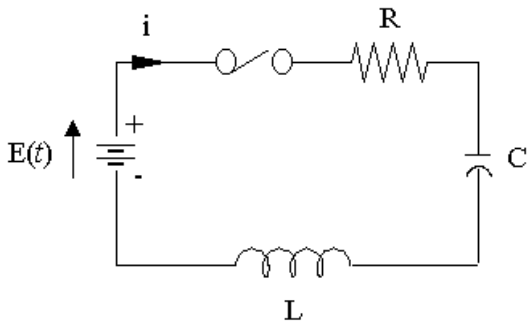

Figure 1: RLC series circuit [11]
E(t), the voltage source powering the circuit
I, the current admitted through the circuit
R, the effective resistance of the combined load, source, and components
L, the inductance of the inductor component
C, the capacitance of the capacitor component

In this circuit, the three components are all in series with the voltage source. The governing differential equation can be found by substituting into Kirchhoff's voltage law (KVL) the constitutive equation for each of the three elements. From the KVL,

$$V_{\mathrm{R}} + V_{\mathrm{L}} + V_{\mathrm{C}} \;=\; V(t) \; or \; E(t) \tag{1.1}$$

where V_R, V_L and V_C are the voltages across R, L, and C, respectively, and V(t) or E(t) is the time-varying voltage from the source.

Substituting $V_{\mathrm{R}} = RI(t), V_{\mathrm{L}} = L\frac{\mathrm{d}I(t)}{\mathrm{d}t}$ and $V_{\mathrm{C}} = V(0) + \frac{1}{C}\int_0^t I(\tau)\mathrm{d}\,\tau$, into the equation above yields: [2]

$$R\,I(t) \;+\; L\frac{\mathrm{d}I(t)}{\mathrm{d}t} \;+\; V(0) + \frac{1}{C}\int_0^t I(\tau)\mathrm{d}\,\tau \;=\; \mathrm{V(t)} \tag{1.2}$$

For the case where the source is an unchanging voltage, taking the time derivative and dividing by L leads to the following second order differential equation:

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}I(t) + \frac{R}{L}\frac{\mathrm{d}}{\mathrm{d}t}I(t) + \frac{1}{LC}I(t) = 0 \tag{1.3}$$

This can usefully be expressed in a more generally applicable form:

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}I(t) + 2\alpha\frac{\mathrm{d}}{\mathrm{d}t}I(t) + \omega_0^2 I(t) = 0 \tag{1.4}$$

α and ω_0 are both in units of angular frequency. α is called the neper frequency expressed in nepers per second, or attenuation, and is a measure of how fast the transient response of the circuit will die away after the stimulus has been removed. Neper is a logarithmic unit of attenuation. ω_0 is the angular resonance frequency expressed in rad/sec.[2]

For the case of the series RLC circuit these two parameters are given by: [3]

$$\alpha = \frac{R}{2L}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \tag{1.5}$$

A useful parameter is the damping factor, ζ, which is defined as the ratio of these two; although, sometimes ζ is not used, and α is referred to as damping factor instead; hence requiring careful specification of one's use of that term.[3]

$$\zeta \equiv \frac{\alpha}{\omega_0} \tag{1.6}$$

In the case of the series RLC circuit, the damping factor is given by

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} \tag{1.7}$$

The value of the damping factor determines the type of transient that the circuit will exhibit. [4]

Transient analysis

The differential equation for the circuit solves in three different ways depending on the value of ζ. These are overdamped ($\zeta > 1$), underdamped ($\zeta < 1$), and critically damped ($\zeta = 1$).

The initial values and the first derivative are related as

$$Ri(0) + L\frac{di(0)}{dt} + V_0, \qquad \frac{di(0)}{dt} = \frac{-1}{L}(RI_0 + V_0)$$

Look for the solution of the form $i = Ae^{S}t$, where A and S are constants. Substituting in differential equation to get

$$S^2 + \frac{R}{L}S + \frac{1}{LC} = 0 \tag{1.8}$$

This quadratic equation is known as the characteristics eqn.

The roots of the equation dictate the character of i and they are given as

$$S_1 = \frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -\omega_0\left(\zeta - \sqrt{\zeta^2 - 1}\right)$$
$$S_2 = \frac{-R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -\omega_0\left(\zeta + \sqrt{\zeta^2 - 1}\right) \tag{1.9}$$

The equation modified in terms of α and ω_0 is $S^2 + 2\alpha S + \omega_0^2$ where, ω_0 is undamped natural frequency.

The two values of S indicate that there are two positive solutions for current i,

$$i_1 = A_1 e^{(S_1 t)}, i_2 = A_2 e^{(S_2 t)}$$

A complete solution would require a linear continuation of $i_1 and i_2$
Thus, the natural response of the series LCR circuit is

$$i(t) = A_1 e^{(S_1 t)} + A_2 e^{(S_2 t)} \tag{1.10}$$

Where the constant A_1 and A_2 are determined from the initial values of $i(0)$ and $\frac{di(0)}{dt}$

Now, we get 3 cases

1. Overdamped Case i.e., $\alpha > \omega_0 \; Or \; R^2 > \frac{4L}{C}$ [5]
   Both the roots S_1 and S_2 are negative and real, the response is given as

$$i(t) = A_1 e^{(S_1 t)} + A_2 e^{(S_2 t)} \tag{1.11}$$

2. Critically damped Case i.e., $\alpha = \omega_0 \; Or \; R^2 = \frac{4L}{C}$ [5]
   We have, $S_1 = S_2 = -\alpha = -\frac{R}{2L}$
   For this case, $i(t) = (A_1 t + A_2)e^{(-\alpha t)}$ (1.12)

3. Underdamped Case i.e., $\alpha < \omega_0 \; Or \; R^2 < \frac{4L}{C}$ [5]
   The natural response is

$$i(t) = e^{(-\alpha t)}\left(A_1 e^{(-j\omega_d t)} + A_2 e^{(-j\omega_d t)}\right) \tag{1.13}$$

   Where, $j = \sqrt{-1}$, and $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \omega_0\sqrt{1 - \zeta^2}$

   Also, $i(t) = e^{(-\alpha t)}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$ (1.14)

The transient response or type that a circuit exhibits is dependent on the value of the damping factor. The damping factor is the amount by which the oscillation of a system gradually decreases with time [2].

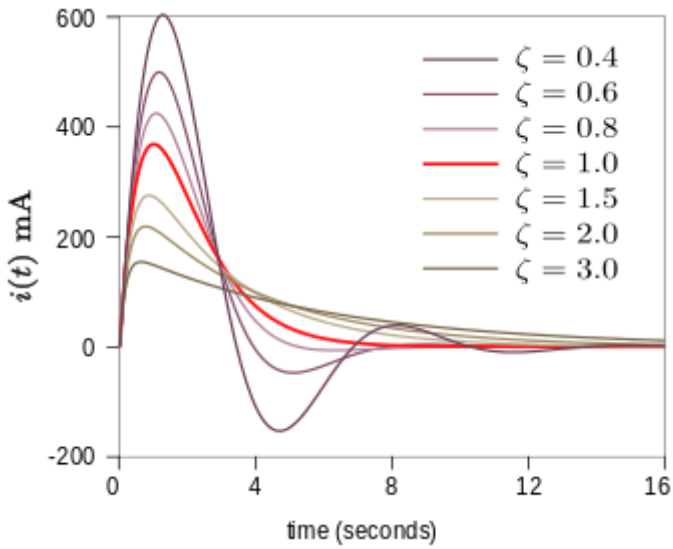

Figure: Plot showing underdamped and overdamped responses of a series RLC circuit to a voltage input step of 1 V. The critical damping plot is the bold red curve. The plots are normalized for L = 1, C = 1 and $\omega_0 = 1$. [12]

Now, as we know that the complete response must satisfy both the initial conditions and the final condition of the forced response i.e., [6]

$$Complete\ response\ =\ Natural\ response\ +\ Forced\ response \tag{1.15}$$

Once we obtain the complete response, the circuit is considered to be solved.

Chapter 2: Objectives

The main objectives of this research are:

a) To solve the 2nd order differential equation of LCR series circuit analytically and using numerical method

b) To analyze the absolute error between analytically solved data and numerical method data

c) To use Runge-Kutta order 4 methods to solve coupled differential equation in FORTRAN

d) To observe the complete response in different numerical cases with the help of plotting software.

Chapter 3: Literature Review

Many works have been done on the solution of differential form of LCR circuit but here we review some of them that are relevant to our work.

Agarwal [3] et al. For a driven LCR series circuit 2nd order differential equation can be written as,

$$\frac{\mathrm{d}^2 V_c(t)}{\mathrm{d}t^2} + \frac{R}{L}\frac{\mathrm{d}V_c(t)}{\mathrm{d}t} + \frac{1}{LC}V_c(t) = \frac{1}{LC}V_{IN}(t) \tag{3.1}$$

Whose total solution will be the sum of the homogenous solution and particular solution as follows,

$$v_C(t) = v_{CH}(t) + v_{CP}(t) \tag{3.2}$$

C. Alexander [5] et al. Determined the step response of series LCR circuit using output voltage i.e., $i = C\frac{dV}{dt}$

So. $\frac{di}{dt} = C\frac{d^2V}{dt}$

We get, Step response

$$\frac{\mathrm{d}^2 V}{\mathrm{d}t^2} + \frac{R}{L}\frac{\mathrm{d}V}{\mathrm{d}t} + \frac{V}{LC} = \frac{V_S}{LC} \tag{3.3}$$

The solution of (3.3) has two components, the transient response $V_t(t)$ and the steady state response $V_{ss}(t)$, that is

$$v(t) = v_t(t) + v_{ss}(t) \tag{3.4}$$

The transient response $v_t(t)$ is the component of the total response that dies out with time. The form of the transient response is the same as the form of the solution for the source-free circuit.

Dorf [6] et al. Also determined the complete response as the sum of natural response and forced response.

$$\boldsymbol{x} = \boldsymbol{x_n} + \boldsymbol{x_f} \tag{3.5}$$

Trench [7] et al. Compared LCR circuit as an electrical analog to spring-mass system with damping. For spring-mass system,

$$my'' + cy' + ky = F(t) \tag{3.6}$$

Chapter 4: Theoretical Framework

The theoretical frameworks include all the parts of term related to this research. Some of the basic and fundamental theoretical terms related to this work are discussed below.

Electric Circuits and Current:[6]

Interconnection of electrical element linked together in a closed path, where electric current may flow continuously is called an electric circuit or electric network.

Electric current may be taken as flow of charge. Hence, current is the time rate of change of charge past a given point. Intrinsic property of matter responsible for electric phenomena is charge. The quantity of charge q can be expressed in terms of the charge of one electron, which is $1.6x10^{-19}C$.

Thus, -1C is the charge of $6.24x10^{1}8$ electrons. So, electric current passing through a specified area is defined by the electric charge passing through the area per unit of time.

Then, we can express current as

$$i = dq/dt \tag{4.1}$$

The unit of current is the ampere (A); an ampere is 1 coulomb per second.

A direct current (dc) is a current of constant magnitude.

If the charge q is known, the current i is readily found using Eq. (4.1). Alternatively, if the current i is known, the charge q is readily calculated. We obtain

$$q = \int_{-\infty}^{t} i\, d\tau = \int_{0}^{t} i\, d\tau + q(0) \tag{4.2}$$

where q(0) is the charge at t = 0 .

Voltage (V): [6]

Work (energy) required to move a unit positive charge from the negative terminal to the positive terminal is the voltage across an element.

The equation for the voltage across the element is

$$v = dw/dq \tag{4.3}$$

Where, v is voltage, w is energy (or work), and q is charge. A charge of 1C delivers an energy of 1J as it moves through a voltage of 1V.

Inductor (L): [6]

The ability of a device to store energy in the form of a magnetic field is known as inductance.

We find that the voltage v(t) across the coil is proportional to the rate of change of the coil current. That is

$$v(t) = L\, di(t)/dt \tag{4.4}$$

Where, the constant of proportionality is L, the inductance of the inductor.

Capacitor (C): [6]

The charge holding capacity of the body is capacitance and the bodies especially designed for storing charges are called capacitor.

A capacitor essentially consists of two conducting surfaces separated by an insulating material called dielectric. The conducting surfaces are called plates of the capacitor and the dielectric are mica, air, paper etc.

The capacitance of a capacitor is proportional to the dielectric constant and surface area of the plates and is inversely proportional to the distance between the plates. In other words, the capacitance C of a capacitor is given by

$$C = \frac{\epsilon A}{d} \tag{4.5}$$

The SI unit of capacitance is coulomb per unit volt which is also called Farad, F. Practical capacitors have capacitances in the order of microfarad or picofarad.

The charge stored by the capacitor is proportional to the capacitor voltage v(t). Thus,

$$q(t) = Cv(t) \tag{4.6}$$

Capacitance is a measure of the ability of a device to store energy in the form of separated charge or an electric field.

The variation of the capacitor charge with respect to time implies a capacitor current i(t), given by

$$i(t) = \frac{d}{dt}q(t) = C\frac{d}{dt}v(t) \tag{4.7}$$

Resistor (R): [6]

The ability of a conductor to oppose the flow of charge through it is known as the resistance of the conductor.

It is measured by the ratio of potential difference V across its ends to the current I flowing through it.

$$R = V/I \tag{4.8}$$

Since, Ohm's Law states that the current flowing through a conductor is directly proportional to the potential difference across its ends, provided that the physical conditions (temperature, mechanical strain, etc.) of the conductor remain the same. was published in 1827.

$$V \alpha I$$
$$V = RI \tag{4.9}$$

Unit of R is ohm, Ω in SI-units.

Resonance:

An important property of LCR circuit is its ability to resonate at a specific frequency, the resonance frequency, f_0. Frequencies are measured in units of hertz. For mathematical convenient, angular frequency, ω0, is used. This is measured in radians per second. They are related to each other by a simple proportion,

$$\omega_0 = 2\pi f_0 \tag{4.10}$$

Natural frequency:

The resonance frequency is defined in terms of the impedance presented to a driving source. It is still possible for the circuit to carry on oscillating (for a time) after the driving source has been removed or it is subjected to a step-in voltage (including a step down to zero).

$$\omega_0 = \frac{1}{\sqrt{LC}} \tag{4.11}$$

Damping:

Damping is caused by the resistance in the circuit. It determines whether or not the circuit will resonate naturally (that is, without a driving source). Circuits that will resonate in this way are described as underdamped and those that will not are overdamped. Damping attenuation (symbol α) is measured in nepers per second. However, the unitless damping factor (symbol ζ, zeta) is often a more useful measure, which is related to α by

$$\zeta = \frac{\alpha}{\omega_0} \tag{4.12}$$

The special case of ζ = 1 is called critical damping and represents the case of a circuit that is just on the border of oscillation. It is the minimum damping that can be applied without causing oscillation.

Kirchhoff's law:[6]

In 1847, Gustav Robert Kirchhoff, a professor at the University of Berlin, formulated two important laws that provide the foundation for analysis of electric circuits. These laws are referred to as Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL) in his honor. Kirchhoff's laws are a consequence of conservation of charge and conservation of energy.

<u>Kirchhoff's current law (KCL)</u>: The algebraic sum of the currents into a node at any instant is zero.

<u>Kirchhoff's voltage law (KVL)</u>: The algebraic sum of the voltages around any loop in a circuit is identically zero for all time.

Runge-Kutta Method:

Runge-Kutta methods are single-step methods. However, with multiple stages per step. They are motivated by the dependence of the Taylor methods on the specific IVP. These new methods do not require derivatives of the right-hand side function f in the code, and are therefore general-purpose initial value problem solvers. Runge-Kutta methods are among the most popular ODE solvers. They were first studied by Carle Runge and Martin Kutta around 1900. Modern developments are mostly due to John Butcher in the 1960s.

Gnu plot:

Gnu plot is the widely used plotting program. It is basically command line base program. It can run in different operating system like Linux , UNIX , windows etc. This tool generates 2d or 3d dimensional graph of the different functions, data and data fit . Also, it can give the output result directly on the screen in different formats like, pdf file, jpeg file etc. It was first released in 1986 and this program is written in 'C' programming language.

Chapter 5: Data Set, Methodology

We introduced RK4 method is an effective and widely used method for solving the initial-value problems of differential equations. Runge–Kutta method can be used to construct high order accurate numerical method by functions' self without needing the high order derivatives of functions.

The Runge-Kutta method finds the approximate value of y for a given x. Only first-order ordinary differential equations can be solved by using the Runge Kutta 4th order method. Hence our 2nd order differential equation needs to be changed into 2 first order differential equation.

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0 \tag{5.1}$$

For two general 1st order ODE's,

$$\frac{dy}{dx} = z = f(x, y, z), \qquad \frac{dz}{dx} = g(x, y, z)$$

The 4th order Runge-Kutta formulas for a system of 2 ODE's are:

$$k_0 = hf(x_i, y_i, z_i)$$
$$k_1 = hf\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_0, z_i + \frac{1}{2}l_0\right)$$
$$k_2 = hf\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1, z_i + \frac{1}{2}l_1\right)$$
$$k_3 = hf(x_i + h, y_i + k_2, z_i + l_2) \; and$$

$$l_0 = hg(x_i, y_i, z_i)$$
$$l_1 = hg\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_0, z_i + \frac{1}{2}l_0\right)$$
$$l_2 = hg\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1, z_i + \frac{1}{2}l_1\right)$$
$$l_3 = hg(x_i + h, y_i + k_2, z_i + l_2)$$

$$y_{i+1} = y_i + \frac{1}{6}(k_0 + 2k_1 + 2k_2 + k_3)$$
$$z_{i+1} = z_i + \frac{1}{6}(l_0 + 2l_1 + 2l_2 + l_3)$$

The local truncation error for this method is $O(h^5)$. It is also important to note that the classical fourth-order Runge-Kutta method requires four evaluations of the function f per time step. [13]

Solution of LCR series circuit:

Equation 1.2 is equivalent to,

$$RI + L\frac{dI}{dt} + \frac{1}{C}q = E \tag{5.2}$$

We have 2nd order differential for LCR circuit with source voltage E(t)

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}I(t) + \frac{R}{L}\frac{\mathrm{d}}{\mathrm{d}t}I(t) + \frac{1}{LC}I(t) = E'(t) \tag{5.3}$$

We solve this differential equation with given boundary condition

$$I(0) = I_0,\ I'(0)$$

Similarly, analogous equation for charge of the capacitor $\frac{\mathrm{d}^2Q}{\mathrm{d}t^2} + \frac{R}{L}\frac{\mathrm{dQ}}{\mathrm{d}t} + \frac{1}{LC}Q = E(t)$, with boundary values $Q(0) = Q_0,\ Q'(0) = \frac{dQ}{dt}(0) = I_0$

approaches zero exponentially as t→∞. Thus, all such solutions are transient

To observe the response and transient, in the following numerical examples we choose different values of R and solve it.

5.1 Numerical Case 1 : Non-constant Voltage Source (Underdamped Case)

In an RLC circuit we have L=1 H, R=10 Ω and C=0.0025 F and at t=0, the current and charge is 0 and i′(0)=0.1 A/s. We have a non-constant EMF (electromotive force) of: E=−0.08 cos 2.5t . Solve, for I. [14]

Solution:

As we know, $Complete\ response\ =\ natural\ response\ +\ forced\ response$ i.e., response due to provided source and transient part (E=0). Let's look at natural response of the given case. For this consider there is no EMF (or constant EMF).

In this case, $R^2 = 10^2 = 100$ and $\frac{4L}{C} = \frac{4x1}{0.0025} = 1600$,
So, we have $R^2 < \frac{4L}{C}$ i.e., the given case is of Under-damped condition, from Equation (1.10)

Now, using Equation (5.2) we can write

$$\frac{di}{dt} + 10i + 400q = E \tag{5.1.1}$$

Where, E is a constant.
Differentiating gives

$$\frac{\mathrm{d}^2i}{\mathrm{d}t^2} + 10\frac{\mathrm{di}}{\mathrm{d}t} + 400i = 0 \tag{5.1.2}$$

We are given, $i(0) = 0, i'(0) = 0.1$
The characteristics equation is: $S^2 + 10S + 400 = 0$,
whose solution is $s_1 = -5 + 19.365j,\ s_2 = -5 - 19.365j$

So, $i = e^{(-5t)}(A\,cos\,19.365t + B\,sin\,19.365t)$ using given B.C. and on solving we get

$$i = 0.00516e^{(-5t)}(sin\,19.365t) \qquad (5.1.3)$$

Here is the graph of Natural response from above analytical solution.

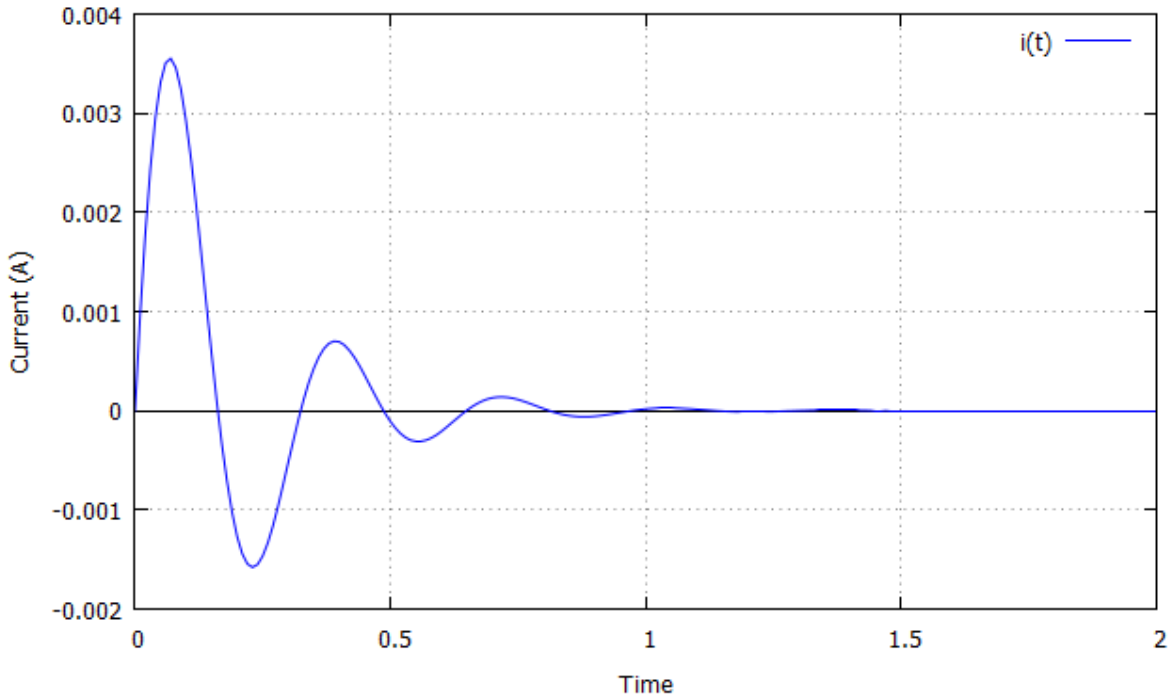

Figure 3 : Graph of natural response for analytical solution, underdamped case

The signal decays very quickly as $t \to \infty$, i.e., the transient response. The initial energy stored in the system dissipates as heat energy in the resistor, initial current is zero. In this solution, the current swings both above and below zero, at least a few times. [15]

Now, solving 5.1.2 using , $i(0) = 0, i'(0) = 0.1$ with R-K4 method.

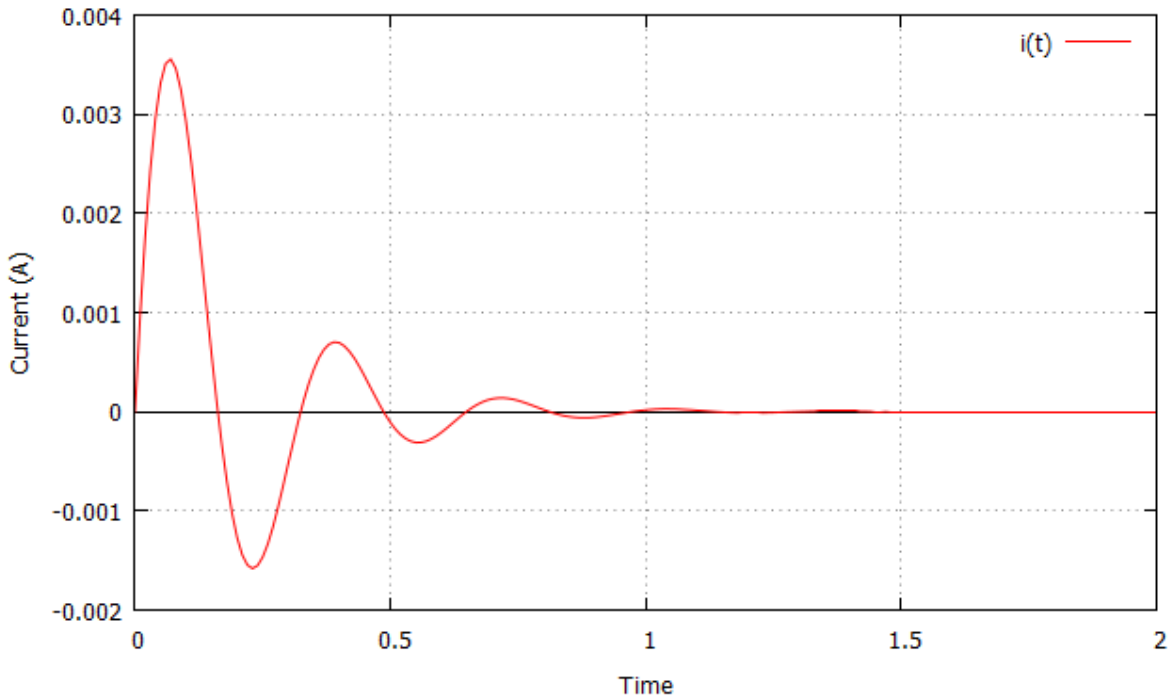

Figure 4: Graph of natural response with RK4 method, underdamped case

Again, using the source values to look at how applied EMF source change the response i.e., Forced response

$$\frac{d^2 i}{dt^2} + 10\frac{di}{dt} + 400i = 0.2\, sin\, 2.5t \qquad (5.1.4)$$

Since, $E = -2.8\, cos\, 2.5t$. Hence, $E' = 0.2\, sin\, 2.5t$
Using B.C. $i(0) = 0, i'(0) = 0.1$

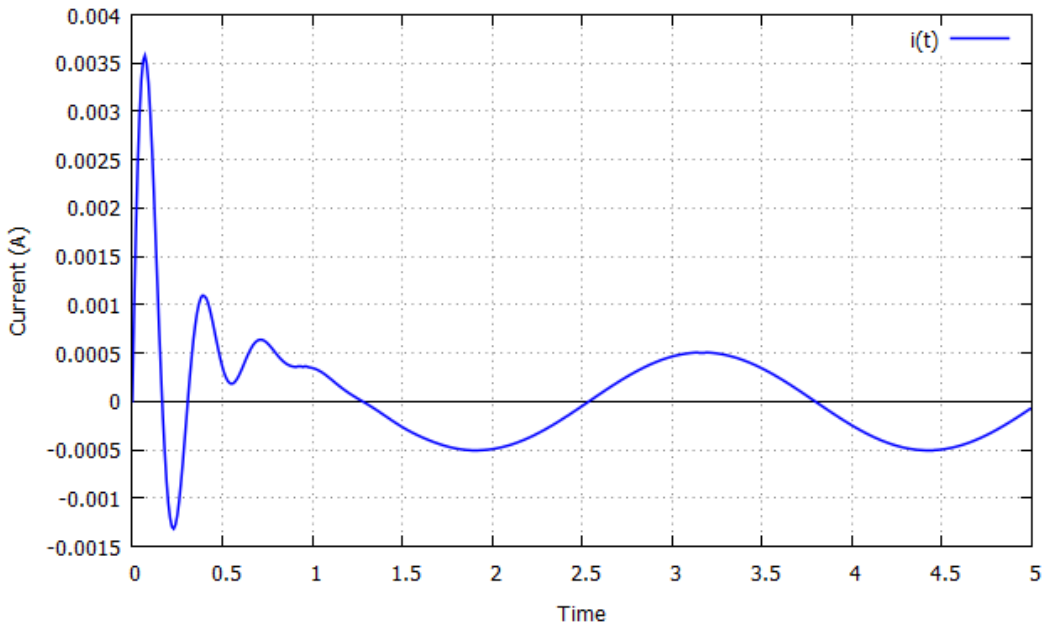

Figure 5: Graph of Forced response obtained through RK4 method, underdamped case

After initial spike current propagates through a sinusoidal pattern.

Numerical Case 2 : Critical damping Case

In an RLC circuit we have L=1 H, R=40 Ω and C=0.0025 F and at t=0, the current and charge is 0 and i′(0)=0.1 A/s. We have a non-constant EMF (electromotive force) of: E=−0.08 cos 2.5t . Solve, for I.

Solution:

EMF source, boundary condition and two energy storage elements of LCR circuit remain same. But resistance is changed from $10 \rightarrow 40$.
For observing the Natural response, we consider there is no EMF (or constant EMF).

In this case, $R^2 = 40^2 = 1600$ and $\frac{4L}{C} = \frac{4x1}{0.0025} = 1600$,
So, we have $R^2 = \frac{4L}{C}$ i.e., the given case is of Critically damped condition, from Equation (1.10)

Now, using Equation (5.2) we can write

$$\frac{di}{dt} + 40i + 400q = E \tag{5.2.1}$$

Where, E is a constant.
Differentiating gives

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + 40\frac{\mathrm{di}}{\mathrm{d}t} + 400i = 0 \tag{5.2.2}$$

We are given, $i(0) = 0, i'(0) = 0.1$
The characteristics equation is: $S^2 + 40S + 400 = 0$,
whose solution is $s_1 = s_2 = -\alpha = -\frac{R}{2L} = -20$

So, $i = i(t) = (A_1 t + A_2)e^{(-20t)}$ using given B.C. and on solving we get

$$i = 0.1t\, e^{-20t} \tag{5.2.3}$$

Now, solving equation 5.2.2 using RK4 method to look at Normal and Forced response.

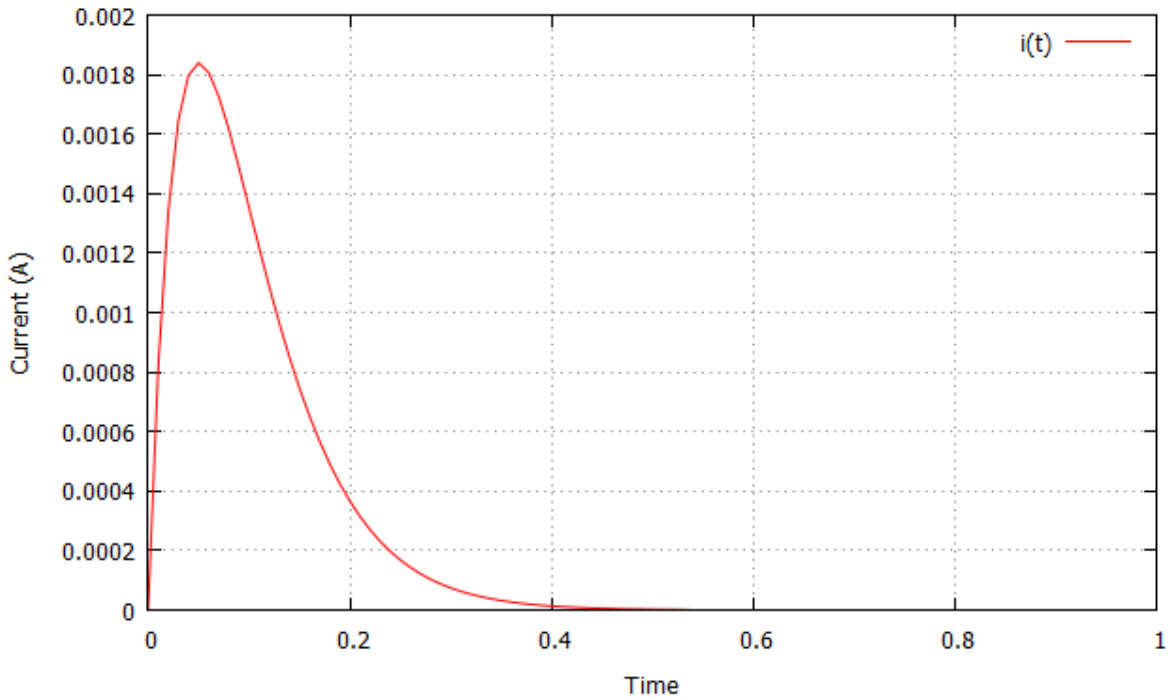

Figure 6: Graph of natural response with RK4 method, critical damping case

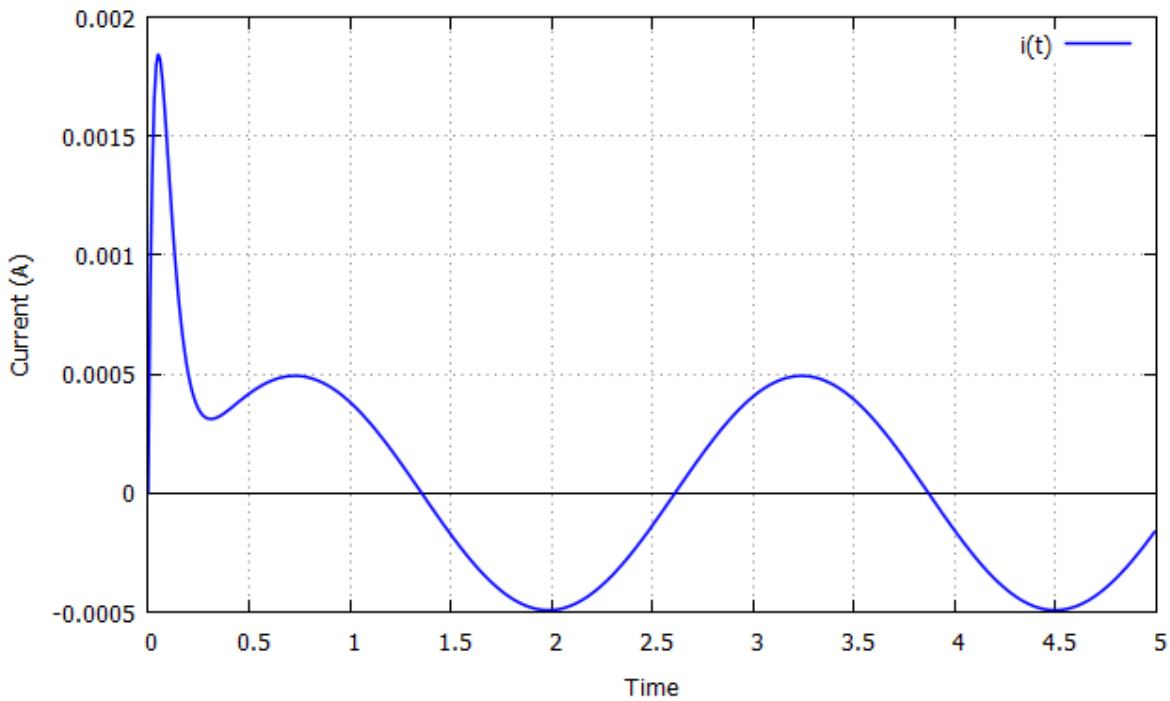

Figure 7: Graph of forced response with RK4 method, critical damping case

5.2 Numerical Case 3 : Overdamped Case

In an RLC circuit we have L=1 H, R=60 Ω and C=0.0025 F and at t=0, the current and charge is 0 and i′(0)=0.1 A/s. We have a non-constant EMF (electromotive force) of: E=−0.08 cos 2.5t . Solve, for I.

Solution:

Once again EMF source, boundary condition and two energy storage elements of LCR circuit remain same. But resistance is changed from $40 \rightarrow 60$.
For observing the Natural response, we consider there is no EMF (or constant EMF).

In this case, $R^2 = 60^2 = 3600$ and $\frac{4L}{C} = \frac{4x1}{0.0025} = 1600$,
So, we have $R^2 > \frac{4L}{C}$ i.e., the given case is of Critically damped condition, from Equation (1.10)

Now, using Equation (5.2) we can write

$$\frac{di}{dt} + 60i + 400q = E \tag{5.3.1}$$

Where, E is a constant.

Differentiating gives

$$\frac{d^2i}{dt^2} + 60\frac{di}{dt} + 400i = 0 \tag{5.3.2}$$

We are given, $i(0) = 0, i'(0) = 0.1$
The characteristics equation is: $S^2 + 60S + 400 = 0$,
whose solution is $s_1 = -30 + 10\sqrt{5},\ s_2 = -30 - 10\sqrt{5}$

So $i(t) = A_1e^{(S_1t)} + A_2e^{(S_2t)}$ using given B.C. and on solving we get

$$i = \frac{0.1}{20\sqrt{5}}e^{(-30+10\sqrt{5})t} - \frac{0.1}{20\sqrt{5}}e^{(-30-10\sqrt{5})t} \tag{5.3.3}$$

Now, solving equation 5.3.2 using RK4 method to look at Normal and Forced response.

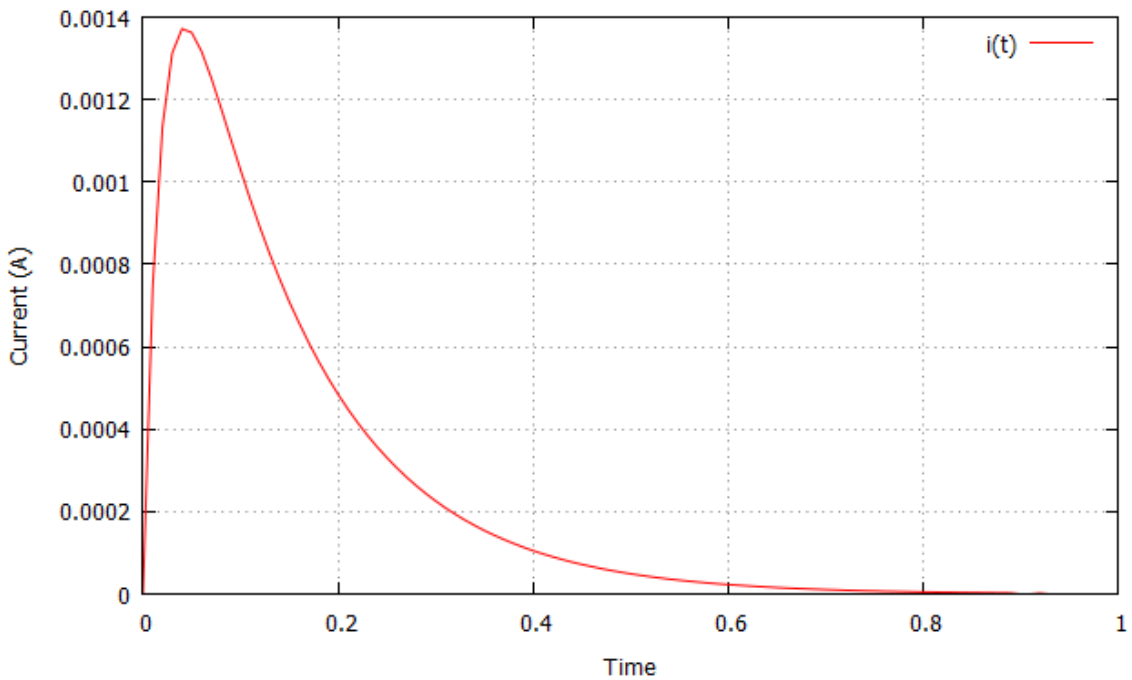

Figure 8: Graph of natural response with RK4 method, overdamping case

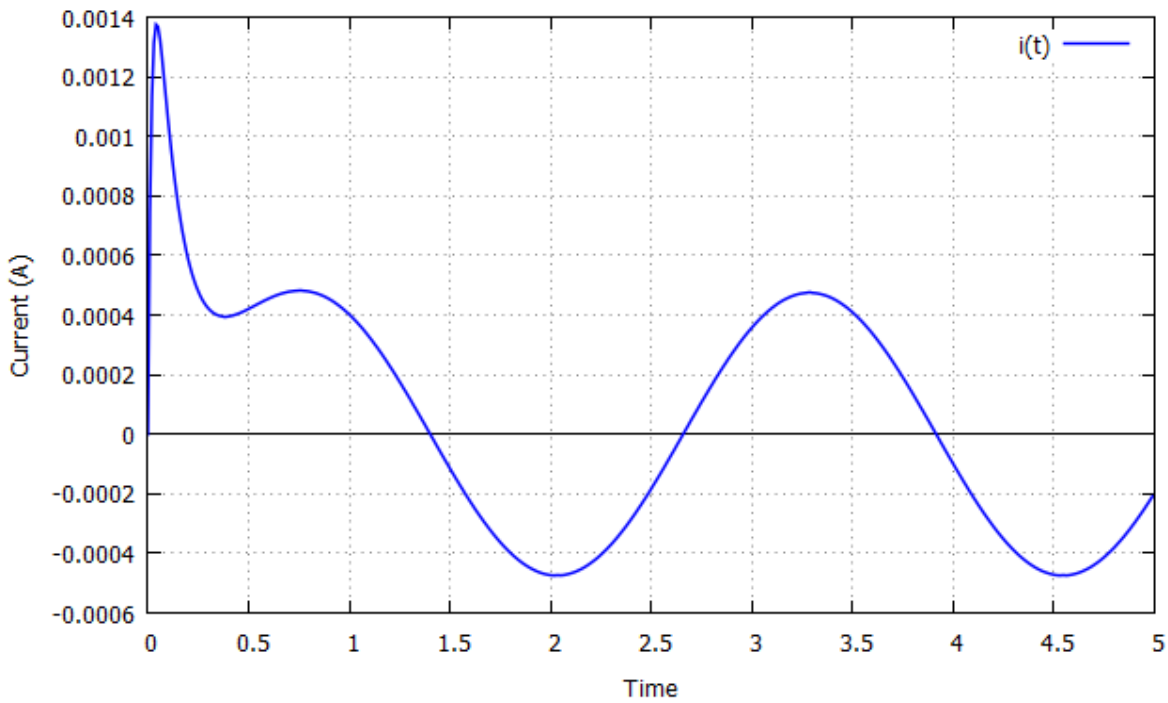

Figure 9: Graph of forced response with RK4 method, overdamping case

Both critical damping and overdamping only have one hump in the wave, in natural response. After initial spike current quickly settles into sinusoidal pattern.

Chapter 6: Result and Discussion

We observed the responses in different numerical cases and found the current analytically & using numerical method i.e.. Runge-Kutta order 4 method. Here, are some results that we obtained.

1. Absolute error between calculated current using Analytical method and Runge-Kutta order 4 method.

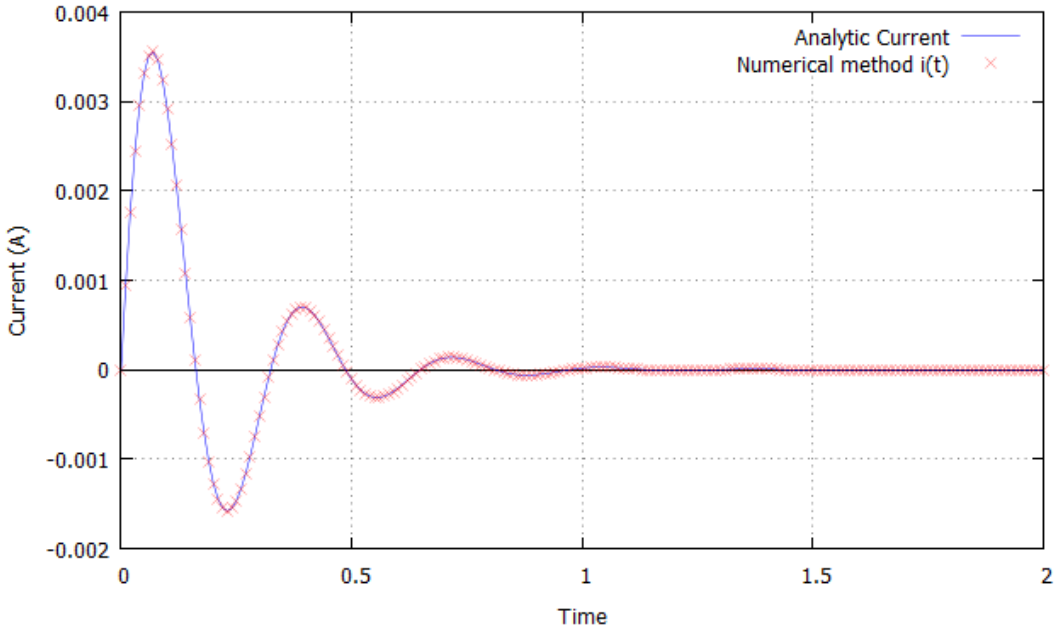

Figure 10: Plot of current data for natural response of both analytical and numerical methods, numerical case 1

Similarly, plotting Absolute Error VS Time, we observe that the error is on the order of 10^{-6} Or less.

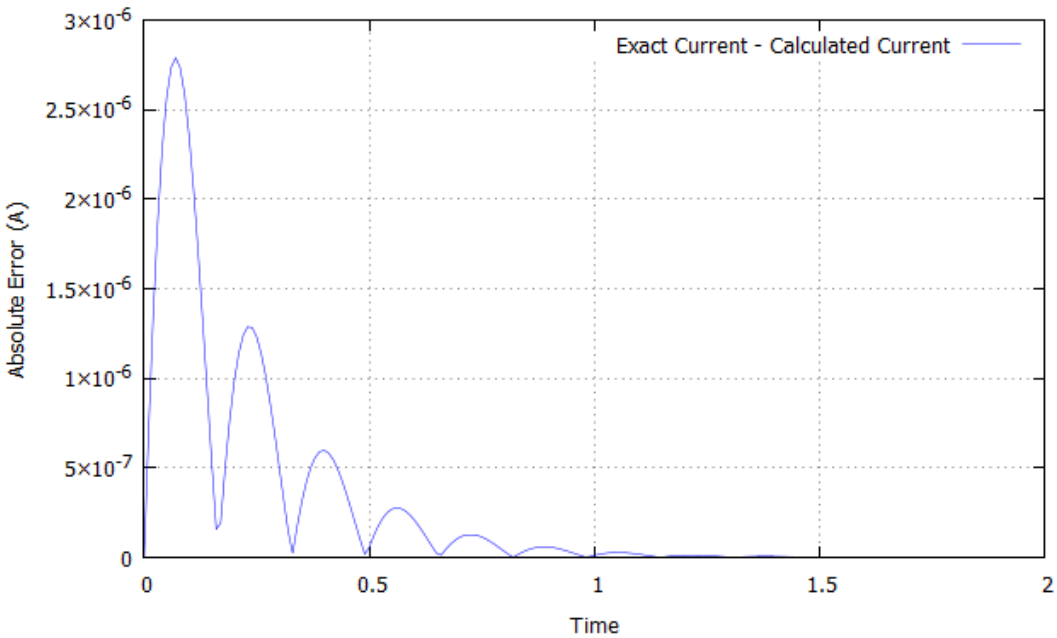

Figure 11: Absolute error V/S Time using data of analytic current and numerical method current for numerical case 1

2. Complete response for the case of Underdamping.

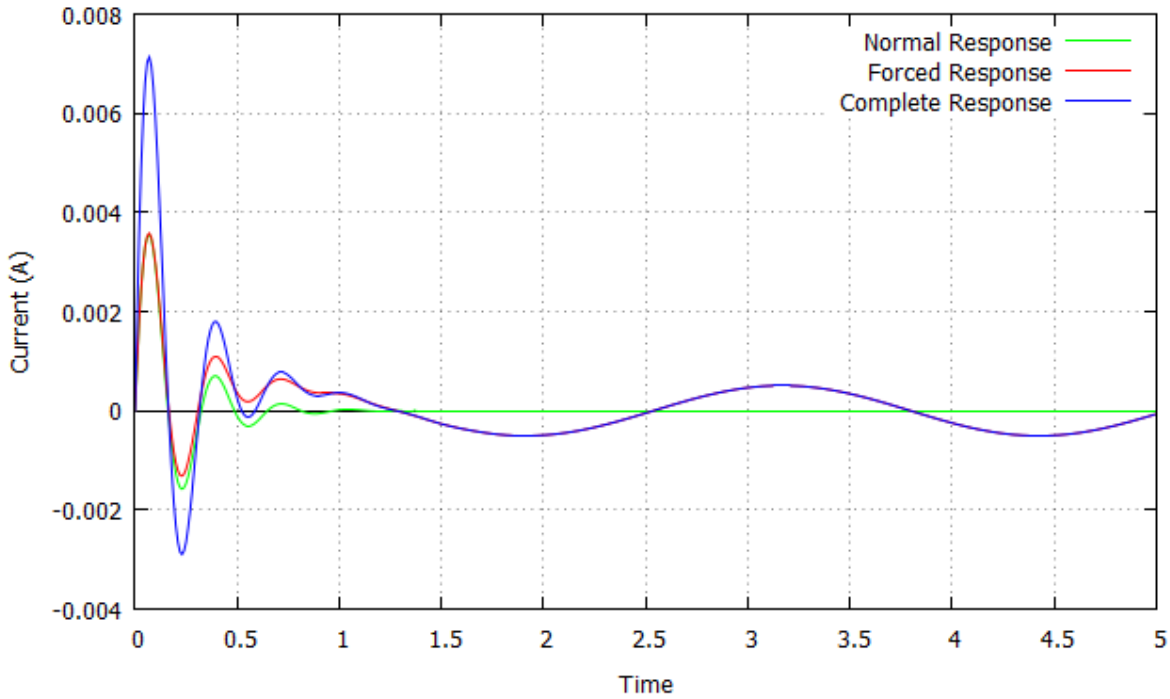

Figure 12: Complete response as a sum of Normal and Forced response for numerical case 1

3. Complete response for the case of Critical damping.

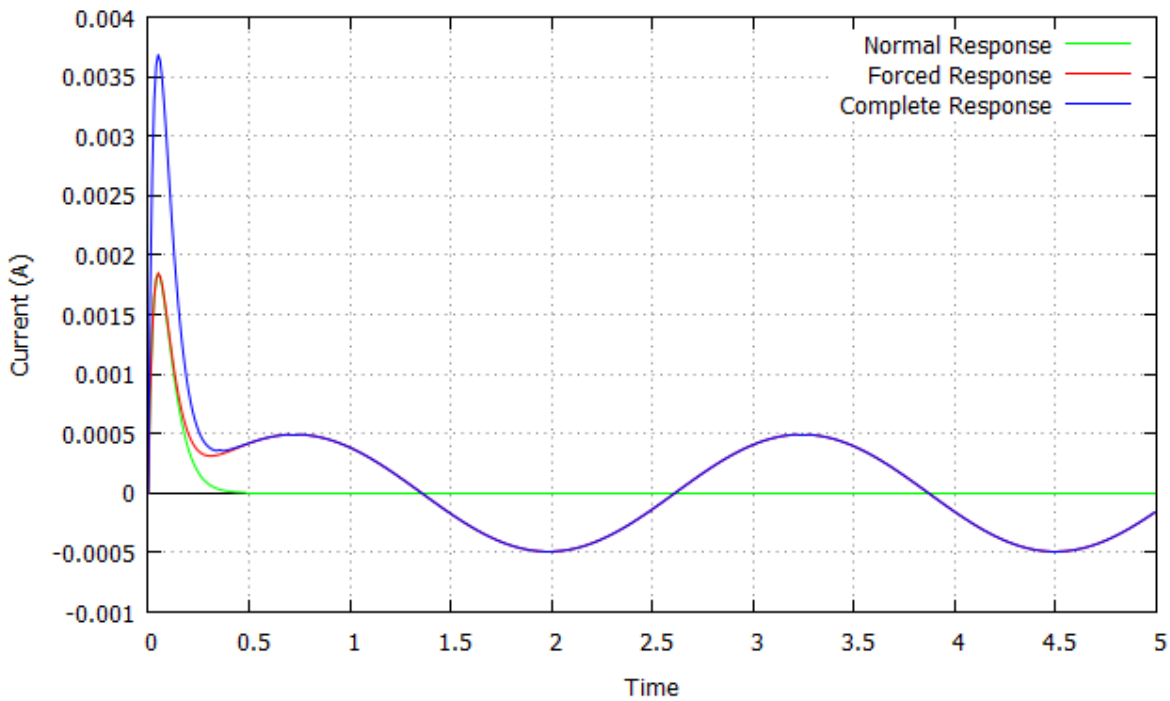

Figure 13: Complete response as a sum of Normal and Forced response for numerical case 2

4. Complete response for the case of Overdamping.

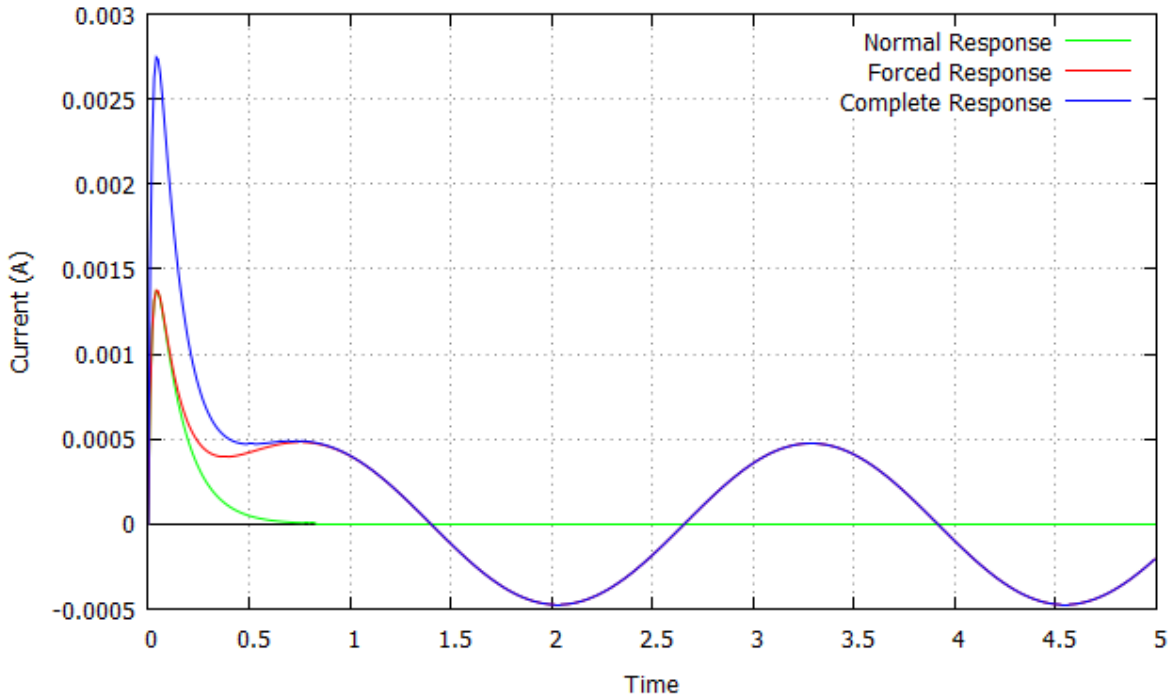

Figure 14: Complete response as a sum of Normal and Forced response for numerical case 3

From Figure 12, 13, 14, it is evident that complete response increases the height of initial spike. Natural responses are transient i.e., they die out after certain time. Wave propagate through a sinusoidal pattern because of AC source (non-constant emf source).

Underdamped current swings above and below zero, at least a few times in natural response. It reaches the highest and wiggles above and below. Both critically damped and over damped current has single hump, i.e., they don't have negative value for current. [15]

Chapter 7: Conclusion & Concluding Remarks

Figure for complete response of 3 cases i.e., Numerical case 1 to 3 with same EMF, same boundary condition and same energy elements (C, L) is shown below:

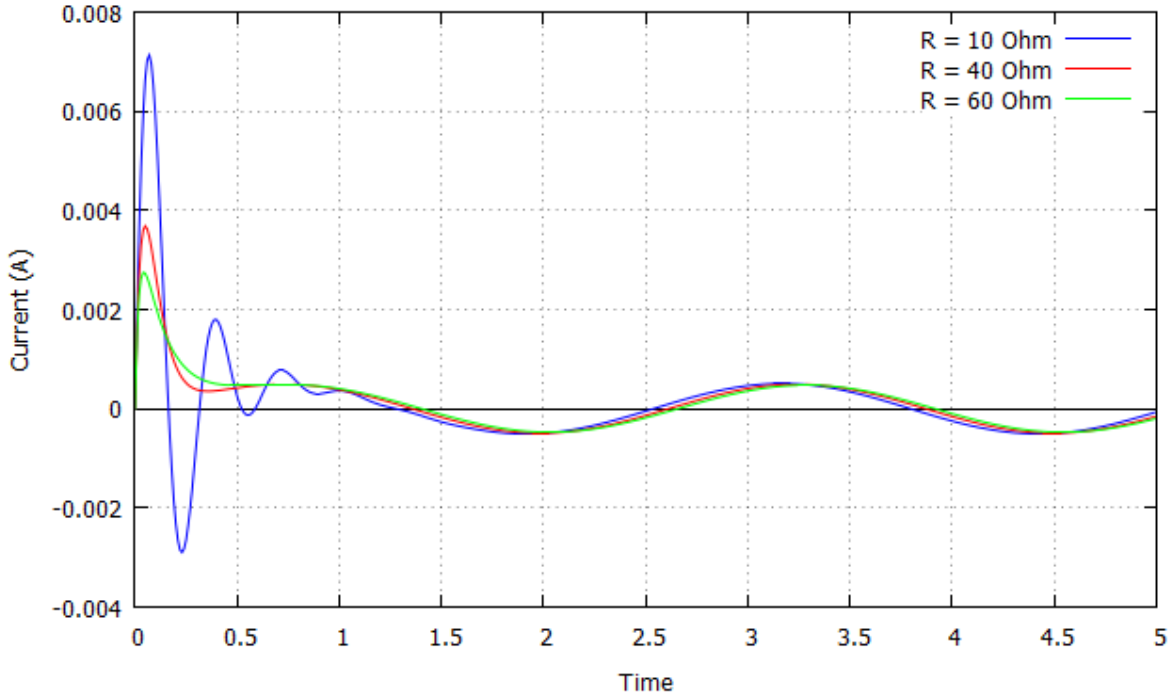

Figure 15: Completer response numerical cases 1, 2, 3

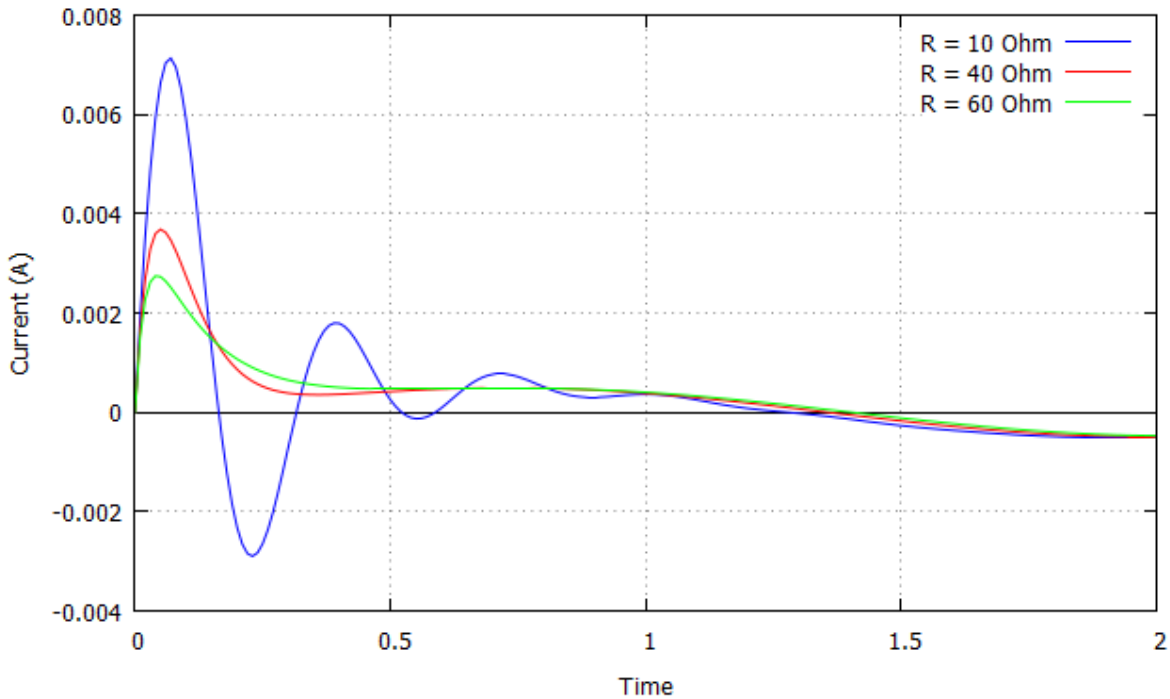

Figure 16: Completer response numerical cases 1, 2, 3 with limit lower than figure 15

The damping effect is due to the presence of resistance R. The damping factor α determines the rate at which the response is damped. By adjusting the value of R, the response may be made underdamped, overdamped, critically damped.

In the case of underdamping, because the storage elements L and C transfer energy back and forth between them we get oscillatory behavior (ringing) (Figure 12). The overdamped has the longest settling time because it takes the longest time to dissipate the initially stored energy. So, for fastest response without oscillation critically damped circuit is the right choice.

Since, our circuit is AC source driven LCR series circuit, critically damped currents settle into sinusoidal pattern quickest (Figure 16). The initial energy in the system dissipates as heat in the resistor. Our solution has initial current as zero.

This paper uses the RK4 method, which is a fourth-order method, meaning that the local truncation error (error caused by one iteration) is on the order of $O(h^5)$, while the total accumulated error (cumulative error caused by many iterations) is on the order of $O(h^4)$. Also, we didn't need to calculate the derivative of f and it's easy for "Automatic Error Control".

Chapter 7: Future works

I have used RK4 method in our project which is more accurate numerical method for solving higher order differential equations when compared to the Heun's method [8] and Euler's method [9] . There are multiple variations of RK4 method which give better result with lower error order, so repeating the project with other variation of RK4, like Butcher RK4 method. Butcher's fifth order Runge-Kutta method is more appropriate and proficient for finding the numerical solutions of initial value problems (IVP) than fourth order Runge-Kutta method. To find more accurate result higher order numerical method is appropriate than lower order methods. [10]

Only one numerical example with same non-constant driving source (emf), same boundary conditions, same energy elements (C, L) and slight variation in R has been used to test currents in different cases. Laboratory experiment to test out our solution and even more variation of R could be done.

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[1] T. L. Floyd, D. L. Buchla *Electronics Fundamentals Circuits, Devices and Applications,* Vol. 8, P. 617, Pearson (2014).

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[3] Agarwal, Anant; Lang, Jeffrey H. *Foundations of Analog and Digital Electronic Circuits*, Morgan Kaufmann (2005).

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[7] Trench, William F., *Elementary Differential Equations*, Trinity University (2013).

[8] Henry, Otoo et al. "Application of Numerical Methods in Transient Analysis." (2019).

[9] Kee, T.A. & Ranom, Rahifa. (2018). Comparison of numerical techniques in solving transient analysis of electrical circuits. ARPN Journal of Engineering and Applied Sciences. 13. 314-320.

[10] Hossain, Md & Alam, & Hossain,. (2020). A Study on Numerical Solutions of Second Order Initial Value Problems (IVP) for Ordinary Differential Equations with Fourth Order and Butcher's Fifth Order Runge-Kutta Methods. 129-137. 10.5923/j.ajcam.20170705.02.

[11] Fig. 1:
https://www.intmath.com/differentialequations/img/HTML/7_2ndODE_damp__1a.png

[12] Fig.2:
https://upload.wikimedia.org/wikipedia/en/thumb/8/8e/RLC_transient_plot.svg/350px-RLC_transient_plot.svg.png

[13] Amzoti, "Help with using the Runge-Kutta 4th order method on a system of 2 first order ODE's.", (2014-03-22): https://math.stackexchange.com/q/721234

[14] Intmath.com, "The Forced Response", https://www.intmath.com/differential-equations/9-2nd-order-de-forced-response.php

[15] Spinning numbers, "RLC natural response – variations", https://spinningnumbers.org/a/rlc-natural-response-variations.html

Appendix:

1. Fortran Programming Implementation:

```
                                <Start of Program>
! Solution of Differential form of LCR circuit using Runge-Kutta Order 4
method
! In an RLC circuit for given L, R & C and given boundary condition.
! ddi/dtt + R/L di/dt + 1/c i = E'/L, di/dt=z=f(t, i, z), dz/dt=ddi/dtt=E'-
(R/L)z-(1/LC)i=g(t, i, z)
! ddy/dxx + R/L dy/dx + 1/c y = E'/L, dy/dx=z=f(x, y, z), dz/dx=ddy/dxx=E'-
(R/L)z-(1/LC)y=g(x, y, z)
! Coded by: Ram Chandra Giri (Author)

!Main Program
program RK4_second_order
    implicit none
    !Variables declaration
    real:: h, a, b, x0, y0, z0
    integer:: i, n
    !Functions declaration
    real:: f, g
    !RK slope
    real:: k1, k2, k3, k4, l1, l2, l3, l4, k, l
    ! dynamic array declaration
    real, ALLOCATABLE :: x(:), y(:), z(:)

    !Varible Initialization
    a=0 !Initial time (sec)
    b=5 !Final Time
    h=0.01 !Step-size (ms) depending on no. of points needed
    n=int((b-a)/h) !No. of steps

    !Array allocation
    ALLOCATE(x(n), y(n), z(n))

    !Creating a file to put data
    open(unit=10, file="data.dat")

    !Boundary condition (Initial Values)
    x0 = a !time (t)
    y0 = _ !current (i) or charge (q)
    z0 = _ !dummy variable, derivative of i/q

    !Array Initial values
    x(1) = x0
    y(1) = y0
    z(1) = z0

    !RK4 part
    do i = 1, n, 1
        k1 = h* f(x(i), y(i), z(i))
        l1 = h* g(x(i), y(i), z(i))

        k2 = h* f(x(i)+h/2, y(i)+k1/2, z(i)+l1/2)
        l2 = h* g(x(i)+h/2, y(i)+k1/2, z(i)+l1/2)

        k3 = h* f(x(i)+h/2, y(i)+k2/2, z(i)+l2/2)
        l3 = h* g(x(i)+h/2, y(i)+k2/2, z(i)+l2/2)
```

```
        k4 = h* f(x(i)+h, y(i)+k3, z(i)+l3)
        l4 = h* g(x(i)+h, y(i)+k3, z(i)+l3)

        k = (k1 + 2*k2 + 2*k3 + k4)/6.0
        l = (l1 + 2*l2 + 2*l3 + l4)/6.0

        x(i+1) = x(i) + h
        y(i+1) = y(i) + k
        z(i+1) = z(i) + l

        !Writing in a datafile: S. no., time, current/charge, derivative of
current
        write(10,*) i, x(i), y(i), z(i)
    end do
    !deallocate arrays
    DEALLOCATE(x, y, z)
    !Closing of File
    close(10)
end program RK4_second_order

!external function
real function f(x, y, z)
    real:: x, y, z
    f = z
end function
real function g(x, y, z)
    real:: x, y, z
    real, PARAMETER:: L=_
    real, PARAMETER:: R=_
    real, PARAMETER:: C=_
    g = (E_dot/L)-(R/L)*z-(1/(L*C))*y
end function
                              <End Of Program>
```

2. **GnuPlot plot file:**

```
# Start of file
set xzeroaxis linetype -1
set xtics auto

#set xrange [0:5]
set autoscale

set xl 'Time'
set yl 'Current (A)'

plot 'data_complete.dat' u 2:5 w l lw 1.5 lc 'blue' title 'i(t)'
# End of file
```